

DELOITTE × UNIVERSITY OF MIAMI · CAPSTONE

Miami-Dade Transit Equity Simulator

A decision tool for where a transit dollar buys the most equity.

Daniel Regalado · MSBA 2026

THE PROBLEM

Transit investment decisions are made without equity in the loop.

Planners know where buses run.

Politicians know where the votes are.

Neither knows where a dollar of service buys the most equity.

MIAMI-DADE TODAY

51

census tracts in **Critical** equity priority
of 504 tracts countywide · home to Miami-Dade's most transit-dependent residents

WHAT WE BUILT

One tool. Four questions.

Where are the equity gaps? · Overview + Tract Explorer

What if we intervene? · Simulate Policy

Which policy wins? · Scenario Comparison

Where do we deploy dollars? · Priority Routes + Recommendations

Built on 504 census tracts, 123 bus routes, ~7,000 stops, ~1M stop-events.

An XGBoost model that understands service — wrapped around an index that understands need.

TECHNICAL

XGBoost v3 · cross-validated $R^2 = 0.82$

13 service + demographic features

Proportional-change propagation to equity score

Fixed-cutoff tier reassignment

BUSINESS

Re-scores every tract in under a second

Answers "what if" without new data collection

Uses **APTA / TCRP** elasticities for ridership

Deterministic wait-time math · no black box

DEMO

Live walkthrough

~8 minutes · huggingface.co/spaces/DanielRegaladoCardoso/miami-transit-equity

OVERVIEW → EXPLORER → SIMULATE → COMPARISON → RECOMMENDATIONS → ROUTES

WHAT THE DEMO SHOWED

Three policies, three answers.

PEAK FREQUENCY BOOST

+2 bus/hr

on Critical tracts

Largest wait-time reduction per tract.

RAIL MODAL SHIFT

+10 pp

Metrorail mode share

Largest county-wide ridership lift.

COMBINED STRATEGY

Moves the most tracts out of Critical. **This is the policy we'd recommend.**

A defensible, repeatable method — not a one-off dashboard.

Reusable · the equity composite + simulator pattern ports to any metro with ACS + GTFS

Auditable · every number traces to a model, a cutoff, or a cited elasticity

Client-ready · built to explain to a mayor, not just a data team

Extensible · cost-per-service-hour overlay and live GTFS refresh are next

WHAT THIS TOOL IS NOT

Honesty before scale.

Not a ridership forecaster · elasticity proxy, not farebox data

Not a cost-benefit analyzer · service hours and cost are next sprint

Not a live dashboard · GTFS snapshot March 2026; refresh is quarterly

Stated up front so nobody oversells the tool.

WHAT'S NEXT

From prototype to platform.

Cost layer · operating cost per service hour → impact per dollar

Live feed · quarterly GTFS refresh, automated baseline update

Second metro · prove the method ports — Tampa or Orlando pilot

API · scenario endpoint other Deloitte tools can call

THANK YOU

Questions.

Daniel Regalado · dxr1491@miami.edu

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